SIX EQUIVALENT CHARACTERIZATIONS OF GEOMETRIC NORMALITY FOR A SCHEME OVER A FIELD

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is complete as written and closes the cycle of implications; no step is referred to the reader.

Editorial note. The counted spans carry 2 navigation stubs and no bibliography citation.

Editorial note. The preamble is the chapter's retained minimal preamble (sources/stacks-varieties/retained-preamble.tex lines 1-77), a verbatim excerpt of the native preamble.

Editorial note. The label lemma-geometrically-normal is defined only in varieties.tex, so no cross-chapter restatement had to be weighed for this card.

Allowed standard prerequisites (editorial).

- Geometric normality of a k-algebra. Algebra, Definition algebra-definition-geometrically-normal, invoked at proof lines 2192-2193 and named in the statement. Supplies the algebraic property to which conditions (4) and (5) are translated.
- Base change of schemes as gluing of spectra of tensor products of coordinate rings. Schemes, Section schemes-section-fibre-products, invoked at proof lines 2197-2200. Supplies that normality of X_{k'} follows from normality of the tensor products of the coordinate rings of affine opens.

Source reference. Original definition 0380: sources/stacks-chow/tags.txt, source lines 4219–4219. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original section 01JO: sources/stacks-chow/tags.txt, source lines 2021–2021. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let k be a field. Let X be a scheme over k. The following are equivalent*

(1) *X is geometrically normal,*
(2) *$X_{k'}$ is a normal scheme for every field extension k'/k,*
(3) *$X_{k'}$ is a normal scheme for every finitely generated field extension k'/k,*
(4) *$X_{k'}$ is a normal scheme for every finite purely inseparable field extension k'/k,*
(5) *for every affine open $U \subset X$ the ring $\mathcal{O}_X(U)$ is geometrically normal (see Algebra, Definition 0380), and*

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(6) $X_{k^{perf}}$ *is a normal scheme.*

Proof. Assume (1). Then for every field extension k'/k and every point $x' \in X_{k'}$ the local ring of $X_{k'}$ at x' is normal. By definition this means that $X_{k'}$ is normal. Hence (2).

It is clear that (2) implies (3) implies (4).

Assume (4) and let $U \subset X$ be an affine open subscheme. Then $U_{k'}$ is a normal scheme for any finite purely inseparable extension k'/k (including $k = k'$). This means that $k' \otimes_k \mathcal{O}(U)$ is a normal ring for all finite purely inseparable extensions k'/k. Hence $\mathcal{O}(U)$ is a geometrically normal k-algebra by definition. Hence (4) implies (5).

Assume (5). For any field extension k'/k the base change $X_{k'}$ is gotten by gluing the spectra of the rings $\mathcal{O}_X(U) \otimes_k k'$ where U is affine open in X (see Schemes, Section 01JO). Hence $X_{k'}$ is normal. So (1) holds.

The equivalence of (5) and (6) follows from the definition of geometrically normal algebras and the equivalence (just proved) of (3) and (4). □